Machine-Learning-Based Cell Counting on a 3D-Printed Motorized Microscope Platform

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***Abstract*— This project presents the development of a low-cost, 3D-printed digital microscope that integrates automated image acquisition through the OpenFlexure software ecosystem. During Phase 1, our team focused on constructing the physical microscope, validating mechanical stability, and ensuring full compatibility between the Raspberry Pi, camera module, motorized stage, and the OpenFlexure Web Interface. Using this system, we successfully captured cell images and verified precise motor control for focusing and stage movement. These results establish a reliable foundation for Phase 2, where artificial intelligence and machine learning techniques will be integrated for automated cell counting and advanced analysis tools.**

Keywords—3D printing, cell detection, digital imaging systems, embedded systems, image processing, machine learning, machine vision, microscopy, open-source platform, Raspberry Pi

I. Background

Cell counting, a fundamental in biological research, biomedical education, and clinical diagnostics, allows researchers and professionals to evaluate both the quality and abundance of cells for use in a wide range of applications. To ensure accurate data and analysis of relevant information, such as amounts of cell culture nutrients, plasmid DNA, and transfection agent, it is crucial that cells are carefully counted [1]. Traditionally, cell counting is performed manually using a hemocytometer and trypan blue. However, manual cell counting is often tedious and time-consuming. Additionally, sources of error arise in various areas, such as distortion of cell definition by human perception, and errors in the preparation and load of cell sample into the hemocytometer [2].

Recent advances in technology, low-cost imaging sensors, and embedded computing platforms have enabled the development of accessible, portable, and low-cost microscopes. Specifically, a 3D-printed, low-cost optical microscope can perform cell imaging and reveal intricate cell details by adding a basic camera and Raspberry Pi for image acquisition and lighting control [3].

Furthermore, Artificial Intelligence (AI) and Machine Learning (ML) techniques are a significant and expanding fields that have the potential to assist in automating the process of cell counting. Integrating Cell Counting Machine Learning Operations (CC-MLOps) can significantly accelerate operational improvements, minimize errors, and enhance model reliability. Successful CC-MLOps have been used to automate cell counting of neuronal cells in rodent brain tissues by implementing GitHub integration, Neptune.ai experiment tracking, and Google Cloud Platform's Vertex AI cloud deployment [4].

Despite these advancements, there are still critical gaps in developing and deploying a low-cost microscope system with machine-learning-based cell counting features. There is an increased need for an affordable, accessible, and fully self-contained system that is able to accurately perform automated image acquisition and cell counting..

II. Objectives

A. Overall Objectives for Project

The objective for this project is to develop a 3D-printed microscope following the OpenFlexure High-Resolution Motorized model [5], that integrates artificial intelligence and machine learning algorithms to automate cell counting. Our project aims to provide a low-cost, sustainable, and accessible tool to students, educators, and research professionals of various backgrounds, educational and professional settings.

B. Objectives for Fall Quarter

Objectives for fall quarter included developing the 3D-printed microscope and ensuring full functionality and compatibility with OpenFlexure's Web Application Interface [6]. Additionally, we aimed to create a foundation for winter quarter by laying out the groundwork for the project's machine learning, computer vision, and front-end-user experience components.

C. Objectives for Winter Quarter

The primary objective for winter quarter is to fully integrate automated cell counting features on a custom designed user-interface that allows for saving CSV data, saving images, time-lapse comparison analysis, and AI-assisted movement tracking.

III. Project Setup

A. Software

Our software is implemented with python and its subsequent libraries. Our Phase 1 (basic microscope design) software uses the flashed full desktop Raspbian-OpenFlexure OS and OpenFlexure Connect software which connects to our Raspberry Pi v4 Model B to control the camera and motors of the microscope. OpenCV-Python (PyPI) which will be used for AI development during our Phase 2 software design.

B. Hardware

Our microscope structure is made of a 3D-printed shell. A Raspberry Pi v4 model B acts as the central processing unit using an SD Card flashed with the Raspbian-OpenFlexure OS. There is an added Sangaboard v0.5 connected for 28BYJ-48 5V stepper motors that control and to help control the 2.7mm achromatic doublet lens, LED, and Raspberry Pi Camera Board v2.

IV. Standards Used and Considered

Our project incorporates several hardware, optical, and software standards to ensure a reliable operation of the 3D printed microscope and to support automated ML cell counting. The computer vision pipeline is built on the OpenCV Python (PyPI) standard, a common framework that provides consistent image processing, feature detection, and motion analysis on embedded platforms like our Raspberry Pi.

A. Objective Lens Standard

The microscope's optical system follows the DIN objective lens standard. The standard defines the focal length, tube length, and mechanical threading used in educational and research microscopes. Our 40x DIN compliant lens provides sufficient resolution for observing individual cells. This allows for accurate measurements during use.

B. Camera Interface Standard

Image capture follows the MIPI Alliance CSI-2 camera interface standard, used by the Raspberry Pi Camera v2. This protocol supports high-speed, low latency image transfer directly to the Raspberry Pi. This allows for more stable performance than USB camera alternatives as well as enabling reliable autofocus and real-time ML analysis.

C. Considered Standards

We also evaluated several alternative standards during development. For computer vision, we considered Hogg's AI protocol for evaluating AI in clinical settings but rejected it due to poor performance on complex biological shapes. This led us to choosing the OpenCV-Python standard. In imaging, we considered USB-based camera interfaces used by other OpenFlexure-supported cameras (Logitech C270, Arducam B0196). These cameras used standard fitting for consumers which ended up being slower and inefficient for real-time processing so we chose the CSI-2 standard used by the Raspberry Pi Camera v2. Looking ahead to Phase 2, we are considering additional standards such as TensorFlow Lite quantization to improve ML performance and dataset compatibility.

V. PROTOTYPES

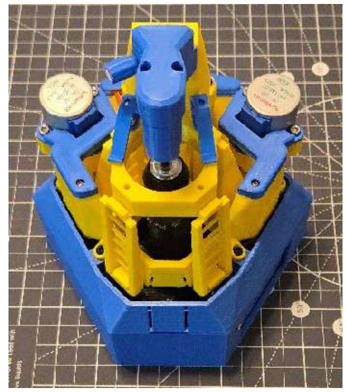

Fig. 1. OpenFlexure microscope build from Phase 1.

The sole prototype of our microscope is an implementation of the OpenFlexure high-resolution motorized microscope in [5] – shown in Figure 1 – serving as both our Fall 2025 quarter product goal and the foundation for our Winter 2026 final product goal. We validated the performance of our prototype by capturing an image from a sample of red onion cells in water – in addition to a successful battery of hardware and software diagnostics.

VI. Experiments and Results

A. Hardware Validation

To evaluate mechanical stability, we performed basic hardware diagnostics after assembly. The 28BYJ-48 stepper motors were tested for smooth and repeatable motion along all three axes. The gear train and lens mount were inspected to confirm proper alignment, and the structural rigidity of the 3D-printed frame was verified to ensure stable focusing without vibration. All motors responded correctly to both manual and automated commands issued through the OpenFlexure interface.

B. Software Functionality Tests

The system was tested using the Raspbian-OpenFlexure OS and OpenFlexure Web Application. Functional tests included live video streaming, autofocus routines, manual stage translation, illumination control, and calibration of the imaging system. The software detected all connected hardware components, confirmed communication between the Raspberry Pi and Sangaboard, and executed movement commands without delay or error.

C. Optical Performancde Test

Fig. 2. Test image of cell walls from a red onion sample.

To verify imaging capability, a red onion epidermal sample was prepared and imaged using the OpenFlexure Web Interface. The microscope produced clear images that demonstrated cellular boundaries and showed consistent illumination. Z-axis adjustments allowed precise focusing with no visible backlash, confirming proper alignment of the optical path and camera module.

D. System Stability and Diagnostics

Long-term stability was evaluated through continuous live imaging, repeated motor movements, and extended illumination tests. Throughout these diagnostics, no thermal issues, mechanical stalls, or software crashes occurred. Captured images were reliably saved to local Raspberry Pi storage, confirming consistent end-to-end operation.

E. Summary of Results

Phase 1 testing demonstrates that the microscope is fully operational, mechanically stable, and compatible with the OpenFlexure software ecosystem. The successful capture of biological sample images and reliable execution of automated control functions validate the platform as ready for Phase 2 integration of AI-based image processing and automated cell-counting features.

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